CRISP-DM Framework: Model Build Phase

DTS 201 - Lecture 7

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Agenda

1. Introduction to Model Build Phase
2. Model Selection Strategy
3. Modeling Workflow
4. Overfitting Prevention
5. Model Assessment Metrics
6. Best Practices
7. Tools and Deliverables
8. Transition to Evaluation
9. Q & A

Where We Are in CRISP-DM

CRISP-DM Phases

1. **Business Understanding**
2. **Data Understanding**
3. **Data Preparation**
4. **MODELING**
5. Evaluation
6. Deployment

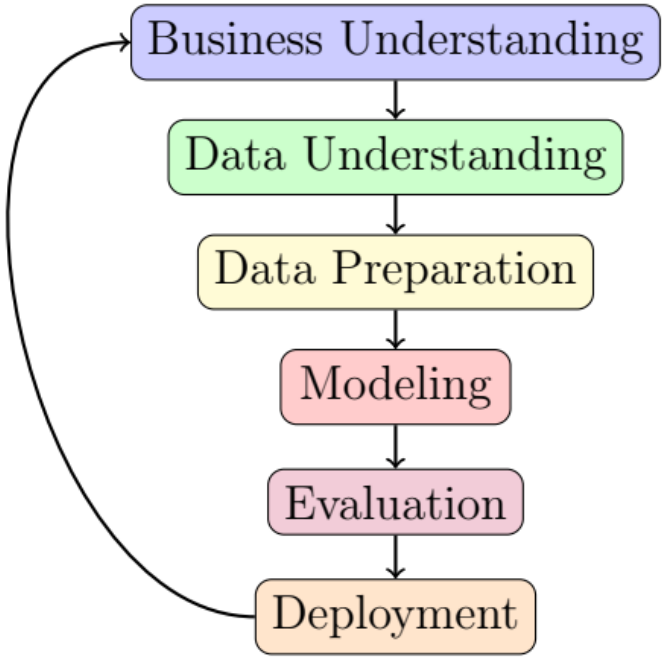


Figure: CRISP-DM Lifecycle

Core Objectives of Model Building

Primary Goals

- Select appropriate modeling techniques
- Build and calibrate models
- Optimize model performance
- Address business objectives

Key Principles

- Iterative experimentation
- Technique-agnostic approach
- Comprehensive documentation
- Business alignment

Matching Problems with Techniques

Problem Type	Common Techniques	Key Considerations
Classification	Logistic Regression, Decision Trees, Random Forest, SVM, Neural Networks	Class imbalance, interpretability needs
Regression	Linear Regression, Decision Trees, Gradient Boosting, Neural Networks	Non-linearity, outlier sensitivity
Clustering	K-Means, DBSCAN, Hierarchical	Cluster count, scalability needs
Time Series	ARIMA, Prophet, LSTM	Seasonality, trend components

Selection Criteria Matrix

Technical Factors

- **Accuracy requirements**
- Training time constraints
- Prediction speed needs
- Data volume/scalability
- Feature types

Business Factors

- **Interpretability needs**
- Deployment environment
- Maintenance requirements
- Regulatory compliance
- Stakeholder preferences

Golden Rule: Start simple, then increase complexity

7-Step Modeling Process

1. Select Techniques
2. Generate Test Design
3. Build Initial Models
4. Preliminary Assessment
5. Tune Hyperparameters
6. Validate Models

Critical Steps Explained

Test Design (Step 2)

Cross-Validation Strategies:

- K-fold (5 or 10)
- Stratified K-fold
- Time-series CV
- Hold-out validation

Hyperparameter Tuning (Step 5)

Methods:

- Grid Search
- Random Search
- Bayesian Optimization
- Evolutionary Algorithms

The Bias-Variance Tradeoff

$$\text{Total Error} = \text{Bias}^2 + \text{Variance} + \text{Irreducible Error}$$

Prevention Strategies

Regularization

- L1/L2 Regularization
- Dropout
- Early Stopping

Validation

- Proper data splits
- Cross-validation
- Out-of-time validation

Simplicity

- Occam's Razor
- Baseline models
- Feature selection

Key Performance Metrics

Classification

- **Accuracy**: $\frac{TP+TN}{Total}$
- **Precision**: $\frac{TP}{TP+FP}$
- **Recall**: $\frac{TP}{TP+FN}$
- **F1-Score**: $2 \times \frac{P \times R}{P+R}$
- **AUC-ROC**

Regression

- **MAE**: $\frac{1}{n} \sum |y - \hat{y}|$
- **MSE**: $\frac{1}{n} \sum (y - \hat{y})^2$
- **RMSE**: $\sqrt{MSE}$
- **R^2**: Variance explained

Essential Do's and Don'ts

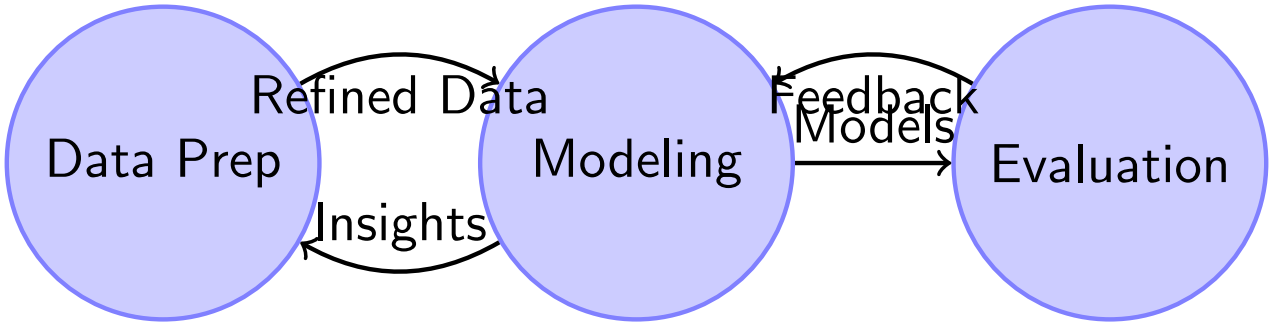
DO: Best Practices

- Start with baseline models
- Use version control
- Set random seeds
- Create pipelines
- Monitor training dynamics
- Involve domain experts

DON'T: Common Pitfalls

- Data leakage
- Over-optimization
- Ignoring baselines
- Premature complexity
- Metric myopia
- Forgetting reproducibility

Iterative Nature of Modeling



Multiple iterations are normal and expected!

Essential Tools Stack

Modeling Libraries

Python:

- scikit-learn
- XGBoost/LightGBM
- TensorFlow/PyTorch

R:

- caret/mlr3
- tidymodels

Experiment Tracking

- MLflow
- Weights & Biases
- TensorBoard
- DVC

AutoML

- H2O
- TPOT
- Auto-sklearn

Key Deliverables

Technical Deliverables

- Trained model files (.pkl, .h5, .joblib)
- Model documentation/cards
- Performance reports
- Feature importance analysis
- Error analysis report
- Reproducible code

Decision Artifacts

- Model comparison matrix
- Trade-off analysis
- Risk assessment
- Deployment readiness report

Readiness Checklist

Technical Readiness	Business Readiness
✓ Models exceed baseline	✓ Business metrics projected
✓ Comprehensive documentation	✓ Stakeholders briefed
✓ Test set evaluation	✓ Trade-offs analyzed
✓ Error analysis complete	✓ Risks identified
✓ Computational specs documented	✓ Next steps planned

Summary: Model Building Success

What Makes This Phase Successful?

- **Systematic experimentation** with multiple techniques
- **Rigorous validation** ensuring generalizability
- **Comprehensive documentation** for reproducibility
- **Business-aware decision making** about trade-offs

Success is NOT measured by:

- Model complexity
- Novel algorithms used
- Academic metrics alone
- Individual preferences

Success = Models that address business problems within constraints

Thank you!